PHYSICS REVISION SHEET (CLASS X)

Sindh Textbook Board Karachi — Core Formulae & Important Derivations

1. Kinematics & Motion

Law / Topic	Formula	Key Variables
First Equation of Motion	$v_f = v_i + at$	v_i = initial velocity, v_f = final velocity
Second Equation of Motion	$S = v_i t + \frac{1}{2}at^2$	S = distance (m), t = time (s)
Third Equation of Motion	$2aS = v_f^2 - v_i^2$	a = acceleration (m/s²)

2. Force, Dynamics & Gravity

- **Newton's 2nd Law:** $F = ma$ (Force in Newtons, N).
- **Law of Gravitation:** $F = G(m_1 m_2) / r^2$ where $G = 6.673 \times 10^{-11}\ Nm^2/kg^2$.
- **Weight vs Mass:** $W = mg$ (where $g \approx 9.8\ m/s^2$ or $10\ m/s^2$).

3. Work, Energy & Power

Quantity	Formula	S.I. Unit
Work Done	$W = F \times d \times \cos(\theta)$	Joule (J)
Kinetic Energy	$K.E. = \frac{1}{2}mv^2$	Joule (J)
Potential Energy	$P.E. = mgh$	Joule (J)
Power	$P = W / t$	Watt (W)

4. Waves & Electromagnetism

- **Wave Speed Equation:** $v = f \lambda$ (Frequency f in Hz, Wavelength λ in m).
- **Ohm's Law:** $V = IR$ (Potential Difference V in Volts, Resistance R in Ω).
- **Electrical Power:** $P = VI = I^2R = V^2 / R$.